

Postconcussion Symptoms After an Uncomplicated Mild Traumatic Brain Injury in Older Adults: Frequency, Risk Factors, and Impact on Quality of Life

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Background: Postconcussion symptoms (PCSs) are common complaints reported by patients after a mild traumatic brain injury (TBI), and these symptoms may lower quality of life. Previous investigations have primarily focused on PCSs in children, adults, and athletes. The frequency, and risk factors, and effects of PCSs for older adults with mild TBIs are unclear. **Purpose:** To investigate the frequency and risk factors of PCSs, and investigate their effects on quality of life over time after mild TBI in older adults. **Methods:** A prospective longitudinal study was performed. All participants were enrolled from the emergency department or neurosurgical outpatient clinics of a medical center. The measurement tools were the Rivermead Post-Concussion Symptoms Questionnaire and the Quality of Life after Traumatic Brain Injury. Measurements were performed on the seventh day, at the first month, and at the sixth month after the head injury. A generalized estimating equation model was used for data analyses. **Results:** One hundred and one older adults (mean age of 76.0 years) with mild TBIs with negative neuroimaging findings were included. Overall, 32.7%, 4%, and 15.8% of the sample reported PCS after 7 days, 1 month, and 6 months of head injury, respectively, revealing a U-shaped trend. We observed that comorbidity measured using the modified Charlson Comorbidity Index was associated with differences in PCSs ($P < .05$). PCSs were an independent predictor of changes in postinjury quality of life ($P < .001$). **Conclusions:** The results indicate that PCS after a mild TBI in older adults is prevalent, even in the chronic phase after a TBI, and PCSs significantly affected the quality of life of our cohort. Therefore, to improve patient quality of life, healthcare providers should employ effective interventions to manage PCSs at different phases after a TBI. **Key words:** mild traumatic brain injury, older adults, postconcussion symptoms, quality of life

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DU TO POPULATION AGING, the epidemiology of traumatic brain injury (TBI) has shifted from younger to older groups. Approximately one-third of TBI survivors are older adults.^{1–3} Among these patients, approximately 70% to 80% are diagnosed as having mild TBI (mTBI). Furthermore, compared with younger adults who have TBIs, older adults with such injuries may experience poorer functional outcomes⁴ and higher mortality.⁵

Postconcussion symptoms (PCSs) are prevalent in adults with mTBI, with rates between 11% and 82% (depending on the diagnostic criteria and the timing of assessment).^{6,7} These symptoms manifest as somatic, cognitive, and affective deficits. These symptoms typically occur within a few days and resolve in a matter of weeks following the head injury, but the symptoms may also persist for months or even years.⁸ PCSs have been associated with inability for return to work⁹ and

impaired health-related quality of life (HRQoL).¹⁰ However, knowledge regarding PCS in older adults is limited; studies have primarily focused mostly on children, adolescents, and young adults.^{11–13} Furthermore, no study has examined the risk factors and consequences of PCSs in older adults. Addressing this knowledge gap regarding PCS in older adults can facilitate early screening of PCSs and planning of effective treatments for mitigating symptoms.

The aims of the present study were (1) to explore the frequency of PCSs in older adults with mTBI; (2) to examine the risk factors for PCSs; and (3) to investigate the effects of PCSs on the HRQoL of older adults with TBI over time. The research hypotheses of the study are that older adults have a high frequency of PCSs on the seventh day, first month, and sixth month after an mTBI and that PCSs are correlated with low HRQoL over time.

METHODS

Study design and participants

In this prospective study, convenience sampling was applied to include participants in the emergency department and neurosurgical clinics of a medical center in northern Taiwan between November 2019 and October 2020. The study was approved by the Taipei Medical University Joint Institutional Review Board (N201909041). Participants who met the following criteria were included in our study: (1) 65 years and older, (2) confirmed diagnosis of a TBI with an initial Glasgow Coma Scale (GCS) score of 13 to 15 with or without loss of consciousness for less than 30 minutes, confusion or disorientation, or other transient neurological abnormalities revealed by computed tomography (CT) after the head injury, and (3) ability to complete the tasks and communicate in Mandarin Chinese or Taiwanese. Complicated mTBI refers to older adults with positive CT findings, whereas participants with negative CT findings were classified as having uncomplicated mTBI. Participants were excluded if they or their family had reported more than 1 TBI prior to the study or if they sustained a second head injury during the study period. Participants diagnosed as having dementia, cognitive impairment, or psychiatric disorders before their head injury or who had a head injury combined with major trauma were also excluded.

Sample size estimation was based on the suggestions of Hedeker for a longitudinal study.¹⁴ We set a 2-tailed α value of .05, a power of 0.8, an attrition rate of 20%, and a measurement time point of 3, resulting in a total sample size of 101.

Measure

Demographic and disease characteristics

An information sheet was used to collect demographic and disease data related to PCSs (including age, gender, and initial GCS score in the emergency department), education level, time since head injury, loss of consciousness, comorbidities, medication history, and cause of head injury.^{6,13} Comorbidities were assessed using the modified Charlson Comorbidity Index (mCCI). The mCCI is a method of predicting mortality by classifying comorbidities. A score of 1, 2, 3, or 6 was assigned to each condition representing the risk of death associated with the condition. The overall score was calculated by summing the individual's scores.¹⁵

Postconcussion symptoms

The Rivermead Post-Concussion Symptoms Questionnaire (RPQ) is a self-administered questionnaire for assessing the severity of PCSs during the last 24 hours.¹⁶ The questionnaire includes items for 16 symptoms categorized in 3 domains: somatic (headaches, dizziness, nausea or vomiting, noise sensitivity, sleep disturbance, fatigue, blurred vision, light sensitivity, and double vision), cognitive (forgetfulness, poor concentration, and taking longer to think), and affective (being irritable, feeling depressed, feeling frustrated, and restlessness). Participants were asked to rate the degree of PCSs within the last 24 hours compared with their preinjury condition using a 5-point Likert-type scale (0 = not experienced at all, 1 = no more of a problem, 2 = a mild problem, 3 = a moderate problem, and 4 = a severe problem). The overall score ranged from 0 to 64; higher scores indicated worse PCSs. The total RPQ score is the sum of all symptom scores but excluding scores of 1. The *International Classification of Diseases, Tenth Revision (ICD-10)* criteria of PCS require a history of TBI and the presence of 3 symptoms or more of the following 8 symptoms: (1) headache; (2) dizziness; (3) fatigue; (4) irritability; (5) insomnia; (6) concentration difficulty; (7) memory difficulty; and (8) intolerance of stress, emotion, or alcohol.¹⁷ Therefore, the current study mapped the symptoms of the *ICD-10* diagnostic criteria on the RPQ. Patients were identified as having PCSs if they had scores of 2 or more for at least 3 of the following symptoms: headache, dizziness, fatigue, irritability, insomnia, concentration difficulty, and memory difficulty.¹⁶

Quality of life

The Quality of Life after Traumatic Brain Injury (QOLIBRI) is a questionnaire that measures HRQoL after TBI.¹⁷ The QOLIBRI comprises 6 subdomains:

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cognition (7 items), self-perception (7 items), daily life and autonomy (7 items), social relationships (6 items), negative emotions (5 items), and physical problems (5 items). Each item was rated on a 5-point Likert-type scale, with 1 indicating *not at all satisfied* and 5 representing *very satisfied*; items for negative emotion and physical problems were reverse-scored, with 1 indicating *very bothered* and 5 indicating *not at all bothered*. The total score for the QOLIBRI was calculated as the average of the scores for each of the 6 domains. For both the total score and subdomain scores, higher scores represent higher HRQoL. The QOLIBRI was translated into the Chinese version in 2014, and has acceptable reliability and validity.¹⁸

Data analysis

All analyses were performed using SPSS 20.0 for Windows (SPSS Statistics, IBM, Armonk, New York). Result with $P < .05$ were considered statistically significant. Descriptive analyses and frequency distributions were used to describe demographics and disease characteristics as well as to identify the frequency of PCSs. Independent t tests and χ^2 tests were used to compare characteristics and frequency of PCSs between patients with and without PCSs. We used a generalized estimating equation (GEE) to examine the risk factors of PCSs and the effects of PCSs on QoL after adjustment for potential confounders over time. Missing data were replaced using the last observations carried forward method. The value of the quasi-likelihood under the independence model

criterion (QIC) was used to select the best working correlation structure.^{19,20}

RESULTS

Participant characteristics

Overall, 101 participants with mTBI were included in the current study. A total of 3 participants withdrew from the study in the sixth month. The primary reasons for withdrawal were sustaining another head injury and lack of time for participation. Table 1 summarizes the demographics and disease characteristics of the study participants. Participants had an average age of 76 years, and the major cause of head injuries was falling (80.2%). All patients were classified as uncomplicated mTBI due to the absence of CT abnormalities. Of the participants, approximately 33% initially experienced PCSs according to their RPQ scores, and approximately 40% of participants initially lost consciousness. A higher percentage of older adults with mTBI experiencing PCSs had hypnotic prescriptions during the study compared with those without PCSs (69.7% vs 47.1%, $P = .03$). The QOLIBRI scores of patients with PCSs were significantly higher than those of the patients without PCSs after 7 days but not after 1 or 6 months.

Distribution and frequency of PCSs

The distributions of the average scores for the RPQ items are presented in Supplementary Material 1

TABLE 1 Demographic and disease characteristics of study participants

Variables	Total ($N = 101$) n (%)	With PCS ($n = 33$) n (%)	Without PCS ($n = 68$) n (%)	P
Age, mean (SD)	76.00 (7.99)	74.39 (7.77)	76.78 (8.04)	.16
Male	43 (42.6)	12 (36.4)	31 (45.6)	.51
Education level below junior high school	59 (58.4)	16 (48.5)	43 (63.2)	.16
Number of medication use, mean (SD)	2.21 (1.51)	2.00 (1.32)	2.31 (1.61)	.34
Hypnotic prescription prior to enrollment	35 (34.7)	15 (45.5)	20 (29.4)	.11
Hypnotic prescription during the study	55 (54.5)	23 (69.7)	32 (47.1)	.03
Analgesic prescription	18 (17.8)	6 (18.2)	12 (17.6)	.94
GCS, mean (SD)	15 (0)	15 (0)	15 (0)	NS
LOC, mean (SD)	40 (39.6)	20 (60.6)	20 (29.4)	.005
mCCI, mean (SD)	4.66 (2.01)	4.52 (2.24)	4.74 (1.91)	.61
Causes of head injury				.44
Fall	81 (80.2)	25 (75.8)	56 (82.4)	
Traffic accident	20 (19.8)	8 (24.2)	12 (17.6)	
Total score of RPQ, mean (SD)	5.95 (4.53)	10.97 (2.64)	3.51 (2.99)	<.001
Total score of RPQ, median (IQR) ^a	6.00 (2-10)	11.0 (9-13)	3 (0-6)	<.001
Total score of QOLIBRI, mean (SD)	71.61 (17.00)	60.91 (12.32)	76.80 (16.59)	<.001

Abbreviations: GCS, Glasgow Coma Scale; IQR, interquartile range; LOC, loss of consciousness; mCCI, modified Charlson Comorbidity Index; NS, no significance; PCS, postconcussion Syndrome; QOLIBRI, Quality of Life after Traumatic Brain Injury; RPQ, Rivermead Post-Concussion Symptoms Questionnaire.

^aData were analyzed using the Mann-Whitney U test.

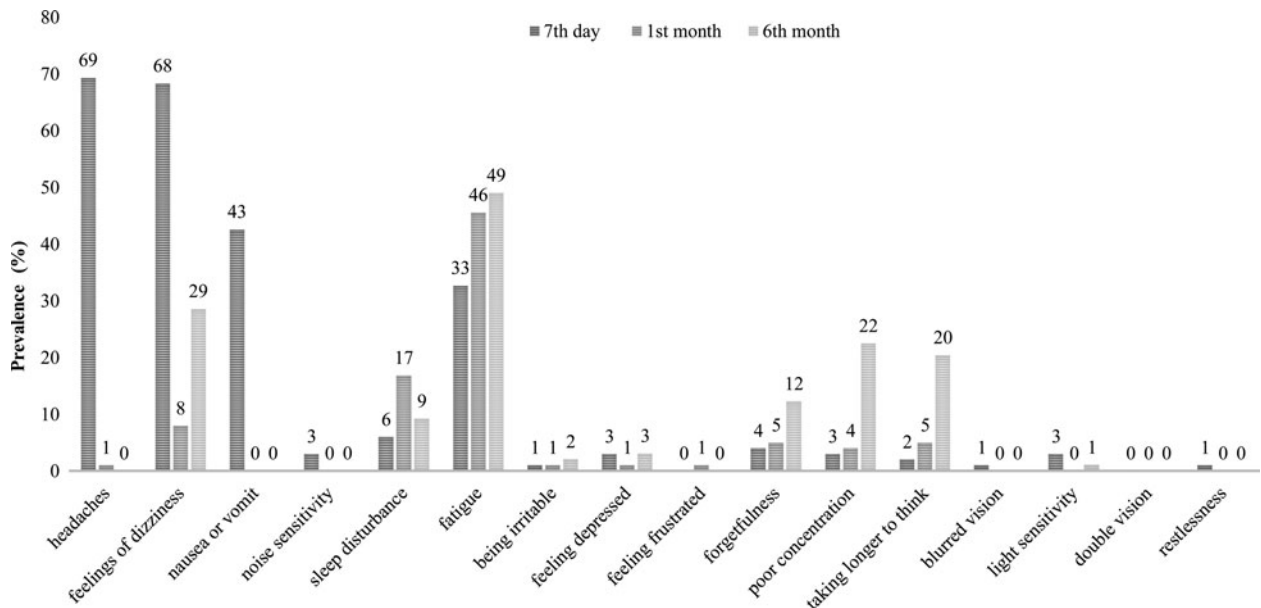


Figure 1. Frequency of postconcussion symptoms at the seventh day, the first month, and the sixth month following traumatic brain injury.

(available at: <http://links.lww.com/JHTR/A479>). Older adults with PCSs had higher total scores of the R PQ than those without PCSs after 7 days, 1 month, and 6 months. As revealed in Figure 1, on the seventh day postinjury, patients with mTBIs reported symptoms. These symptoms were primarily somatic (ie, headaches, dizziness, nausea or vomiting, and fatigue). These somatic symptoms had typically resolved 1-month postinjury. Approximately one-half and one-fifth of older participants with mTBIs experienced fatigue and sleep disturbance, respectively. Fatigue, dizziness, and cognitive deficits (ie, poor concentra-

tion, slower cognition, and forgetfulness) were the major symptoms 6 months postinjury. Overall, the frequency of PCSs on the seventh day, first month, and sixth month after mTBIs was 32.7%, 4%, and 15.8%, respectively.

Risk factors for PCSs in older adults with mTBIs over time

As depicted in Table 2, we observed a U-shaped change in PCS frequency in the GEE findings. After controlling for potential covariates, only mCCI was

TABLE 2 Results of generalized estimating equations in risk factors for PCS in older adults with mild traumatic brain injury over time (N = 101)

Variable	OR	95% CI	P
Intercept	4.83	0.01 to 94.64	.61
Age	0.99	0.93 to 1.06	.83
Male	0.38	0.16 to 0.89	.03
Education level above senior high school	1.29	0.52 to 3.17	.59
mCCI	1.27	1.00 to 1.60	.04
Loss of consciousness	4.57	1.98 to 10.55	<.001
Hypnotic prescription	2.80	1.06 to 7.38	.04
Time			
Sixth month	0.07	0.02 to 0.28	<.001
First month	0.02	0.001 to 0.19	<.001
Seventh day	Reference		
Postinjury sixth month × mCCI ^a	1.39	1.10 to 1.77	.007
Postinjury first month × mCCI ^a	1.32	0.91 to 1.92	.14

Abbreviations: mCCI, modified Charlson Comorbidity Index; mTBI, mild traumatic brain injury; OR, odds ratio; PCS, postconcussion syndrome.

^aData compared with the reference group of postinjury seventh day without taking hypnotic.

associated with changes in PCSs over time, indicating that participants with higher mCCI scores in the sixth month had a greater risk of PCSs than at baseline (odds ratio [OR] = 1.39; 95% CI = 1.10 to 1.77).

Association between changes in PCSs and QoL in older adults with mTBIs

As presented in Table 3, after adjustment for potential covariates, only changes in PCSs were significantly correlated with changes in HRQoL, indicating that PCSs after an mTBI significantly predicted changes in HRQoL over time.

DISCUSSION

To the best of our knowledge, this is the first study confirming that PCSs are prevalent and persist for over 6 months following mTBIs in older adults. Unlike a previous study, which revealed a gradual decline in the frequency of PCSs among young adults after mTBIs,²¹ we observed that the frequency of PCSs was 32.7%, 4%, and 15.8% on the seventh day, first month, and sixth month, respectively, following mTBI, thus presenting a U-shaped trend. Physician visits and medication prescriptions might explain the discrepancy in these findings. In Taiwan, most older adults with mTBIs visited neurosurgeons to confirm the injury and to attend follow-ups for their medical problems in the first month postinjury. These patients might be prescribed medication to treat or manage PCSs, explaining the relative low frequency of PCSs in the first month after head injury.

Nonetheless, the exact reason for the low frequency of PCSs in the first month after an mTBI in older adults could not be ascertained in this study; therefore, further investigations are warranted.

The results revealed that the mean total RPQ score in older adults with PCSs after mTBI was 10.97, which is in accordance with previous literature (mean age = 54 years and mean total RPQ score = 11.31).¹³ However, mean total RPQ scores were much lower than those of a sample of younger patients with mTBI in a study by Smith-Seemiller et al²² (mean age = 38.3 years and mean total RPQ score = 36.3). An explanation for this discrepancy could be differences in the severity of head injury as determined by loss of consciousness and GCS score. Smith-Seemiller et al²² included patients with GCS scores of 13 to 15 and posttraumatic amnesia of less than 24 hours, whereas Ponsford et al¹³ enrolled participants with a GCS score of 15 (95.6%) and primarily without loss of consciousness (72.9%), in line with the characteristics of the patients in our sample. However, further investigation of potential confounding factors is required.

Studies have suggested that somatic domain PCSs (eg, headache and dizziness) are most commonly reported during the acute phase (<1 week postinjury), whereas cognitive and affective domain PCSs are more common during the subacute and chronic postinjury phases for young adults following an mTBI.^{23,24} Our findings suggest a similar phenomenon in the acute phase; however, cognitive symptoms (not affective ones) were dominant in the chronic phase among older adults with mTBIs.

TABLE 3 Results of generalized estimating equations in the association between postconcussion syndrome and health-related quality of life in older adults with mild traumatic brain injury over time (N = 101)

Variables	B	SE	95% CI	P
Intercept	108.38	15.16	78.67 to 138.10	<.001
Age	−0.25	0.22	−0.69 to 0.19	.27
Male	1.13	2.46	−3.68 to 5.95	.65
Education level above senior high school	1.20	2.65	−3.99 to 6.39	.65
Loss of consciousness	1.37	2.24	−3.02 to 5.76	.54
mCCI	−2.96	0.83	−4.60 to −1.33	<.001
Hypnotic prescription during the study	−2.73	2.11	−6.86 to 1.41	.20
Having PCS	−13.07	1.19	−15.41 to −10.74	<.001
Time (postinjury)				
Sixth month	6.61	1.24	4.19 to 9.04	<.001
First month	6.07	0.97	4.17 to 7.97	<.001
Seventh day	Reference			
Postinjury sixth month × having PCS ^a	11.57	1.63	8.37 to 14.76	<.001
Postinjury First month × having PCS ^a	13.89	2.71	8.58 to 19.19	<.001

Abbreviations: mCCI, modified Charlson Comorbidity Index; mTBI, mild traumatic brain injury; PCS, postconcussion syndrome; SE, standard error.

^aReference: Postinjury seventh day × without PCS.

An explanation for this finding is that older adults have greater emotional stability and are happier than young adults, resulting in less severe affective symptoms after an mTBI.^{25,26}

Prior works have produced mixed findings regarding the associations between comorbidity and PCSs. Voormolen and colleagues⁶ reported that comorbidity might reduce the risk of PCS (OR = 0.54). By contrast, 2 previous studies have concluded that the presence of preinjury and concurrent psychological status (eg, anxiety and depression) was considered as significant predictors of PCSs in younger and middle-aged TBI (ORs = 2.99 for preinjury psychological status).^{13,27} Our findings further support that comorbidity may be an important marker of postconcussional symptoms, as we found that increasing mCCI substantially elevated the risk of PCSs (OR = 1.27) and positively predicted changes in PCSs. Nonetheless, due to variability in methods of comorbidity measurement, further studies are warranted to verify our findings.

The association between PCSs and HRQoL in young adults with head trauma is unclear, with some studies finding no link²⁸ and others supporting this association.^{29,30} The results of this study support a positive association. However, participants with PCSs in the first and sixth months after an mTBI reported greater QoL changes than they did at the seventh day. At the seventh day, first month, and sixth month, QOLIBRI scores were 76.80, 82.41, and 84.20, respectively, in patients without PCS and 60.91, 71.68, and 74.97, respectively, in TBI patients with PCSs. Ceiling effects for the manifestation of QoL changes may explain this observation because improvements in QoL over time for TBI patients with PCSs would be greater than for those without PCSs.

Limitations

This study has several limitations. First, because of the advanced age of our patients, their abilities to understand, hear, and express, as assessed using the RPQ, may have resulted in an underestimation of PCSs. Second, the COVID-19 pandemic has negatively affected participants' daily lives; for example, limited outdoor activity and increased anxiety may have inflated PCS severity and QoL changes. Therefore, the effects of the pandemic should be considered when interpreting our findings. Third, because several potential covariates (eg, preinjury psychological issues)¹³ and exact medication dosages were not investigated in the current study, the cohort may not be representative of the mTBI population overall. Fourth, other factors (eg, litigation³¹ and chronic pain²² prior to study) that may have influenced symptom reporting were not evaluated in the current study. Finally, previous studies have demonstrated that self-reported RPQ scores might result in nonspecific PCS endorsement in comparison with open-ended interviews.³² Due to these limitations, investigators should use caution when interpreting these findings.

CONCLUSION

The frequency of PCSs in older adults was 32.7%, 4.0%, and 15.8% on the seventh day, first month, and sixth month, respectively, after mTBI. A significant link was observed between increased mCCI scores and differences in PCSs. Additionally, changes in PCSs independently predict changes in QoL after a TBI. Therefore, our findings provide insight into PCSs after a TBI for healthcare providers, facilitating the planning of effective interventions for managing PCSs at different phases after a TBI.

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